DOCKS Quaternions

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Host institution: CENSUS, Observatoire de Paris

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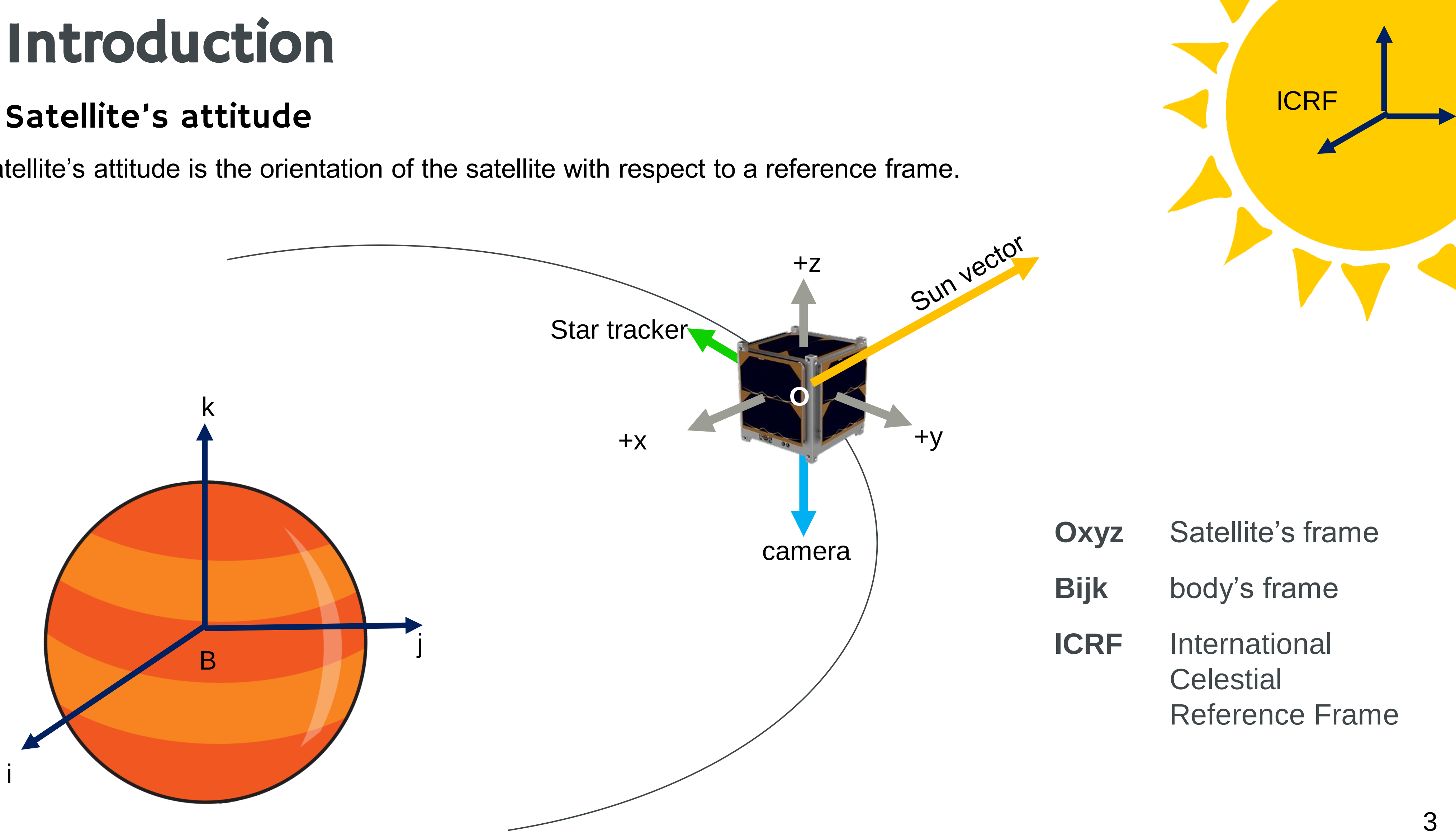
I. Introduction

1. Satellite's attitude

Satellite's attitude is the orientation of the satellite with respect to a reference frame.



Oxyz Satellite's frame

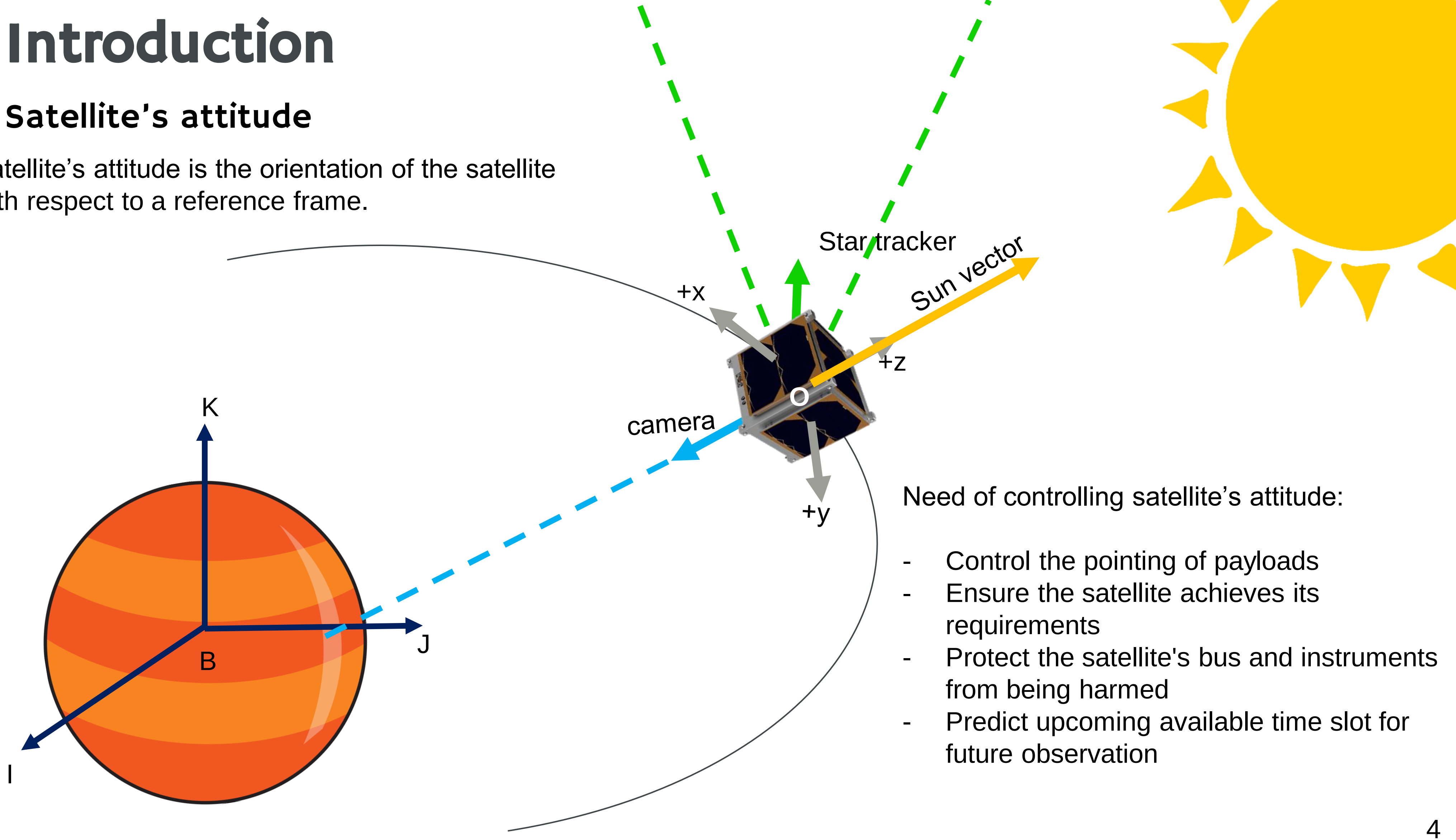
Bijk body's frame

ICRF International Celestial Reference Frame

I. Introduction

1. Satellite's attitude

Satellite's attitude is the orientation of the satellite with respect to a reference frame.



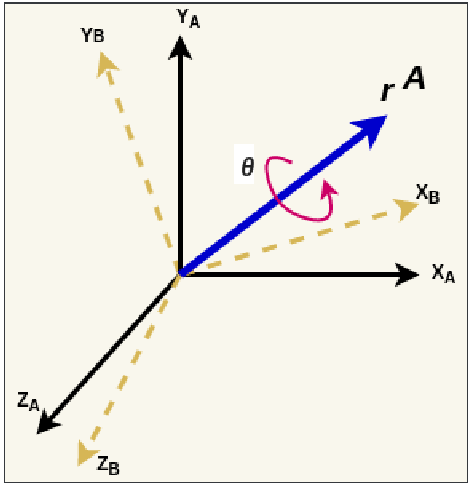
Need of controlling satellite's attitude:

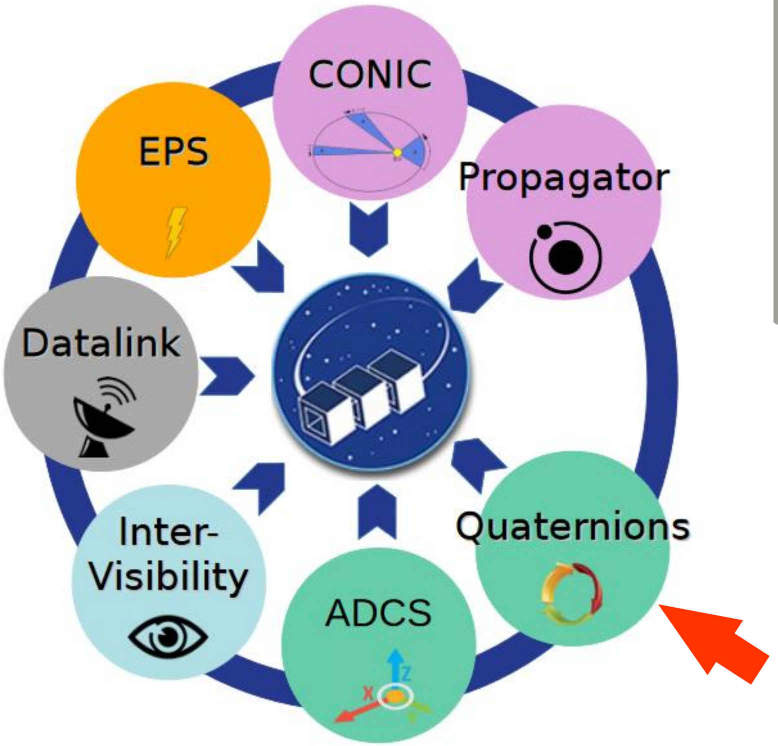
- Control the pointing of payloads
- Ensure the satellite achieves its requirements
- Protect the satellite's bus and instruments from being harmed
- Predict upcoming available time slot for future observation

I. Introduction

2. Quaternions

- Quaternion is a set of four parameters to present the transformation between frames.
- Advantages: Only 4 parameters, no singularities, **good for computation**
- Disadvatages: Hard to visualize





DOCKS

Design and Operation Cross-checking Services

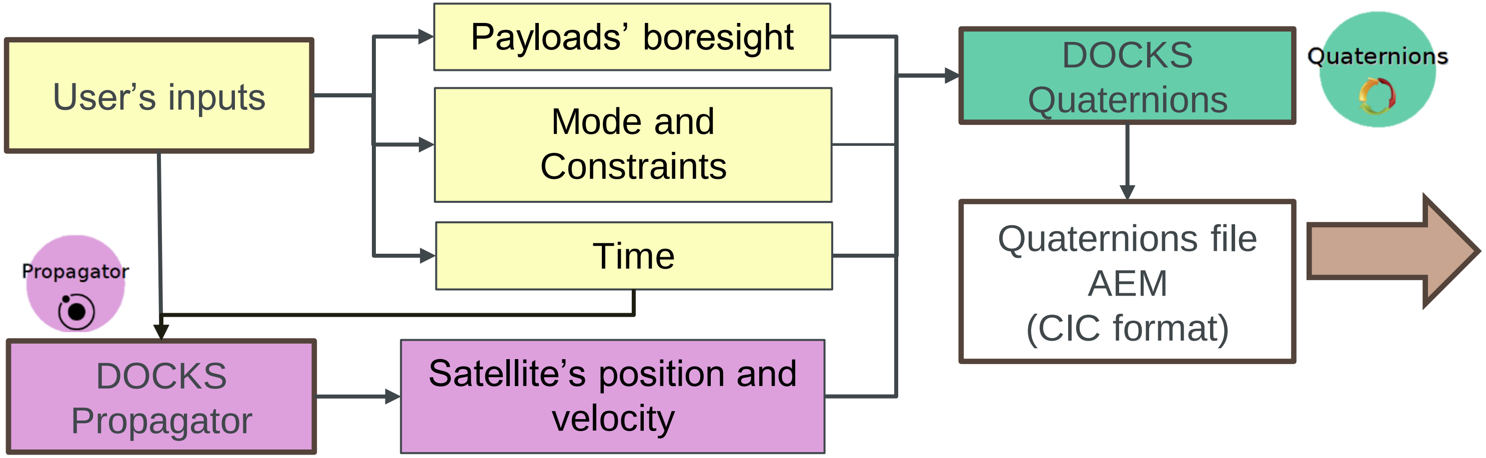
Credit: DOCKS @ CENSUS

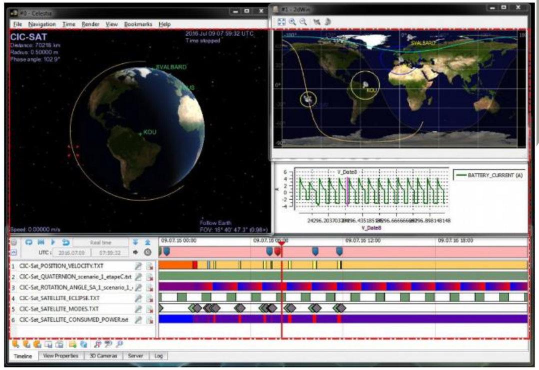
Notes:

- Programming languague: Python
- Abbreviation: LoS – Line of Sight, FOV – Field Of View, pl: payload

II. Scientific objectives

1. Develop pointing stategies (modes and constraints) for satellite.
2. Build the Quaternions module.





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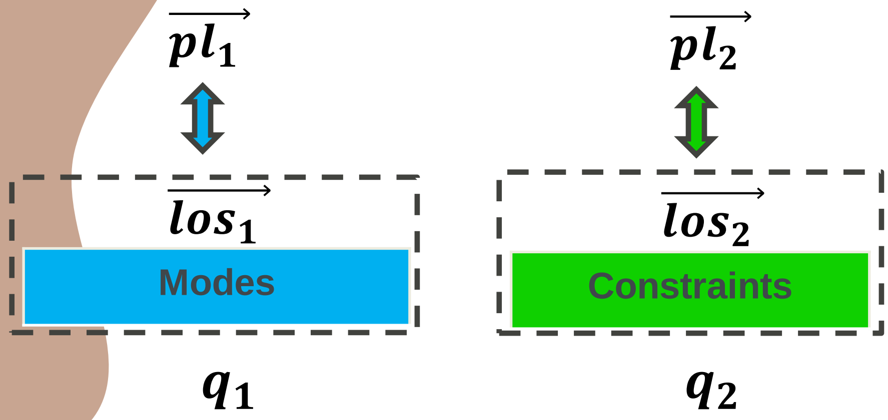
III.Methods

1. How to make pointing strategies?

- Investigation of common pointing strategies
- (A great deal of) Quaternion Algebra
- (A great deal of) Geometry

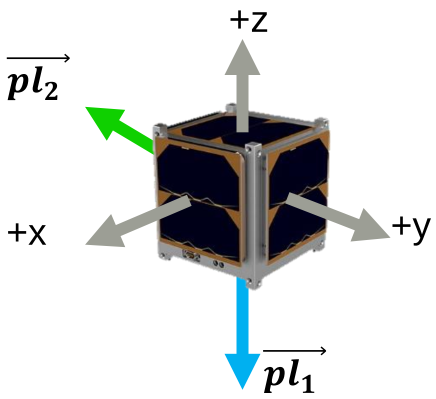


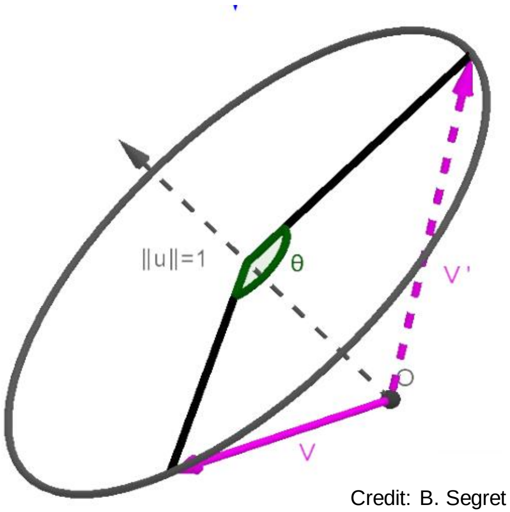
2. How to make a quaternion? It requires:

- Axis of rotation: $\vec{u}$ ($\vec{u}$ is normalized)
- Angle of rotation: θ
- The quaternion is then : $q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right) * \vec{u}$,

  which perform a rotation of θ around vector $\vec{u}$.

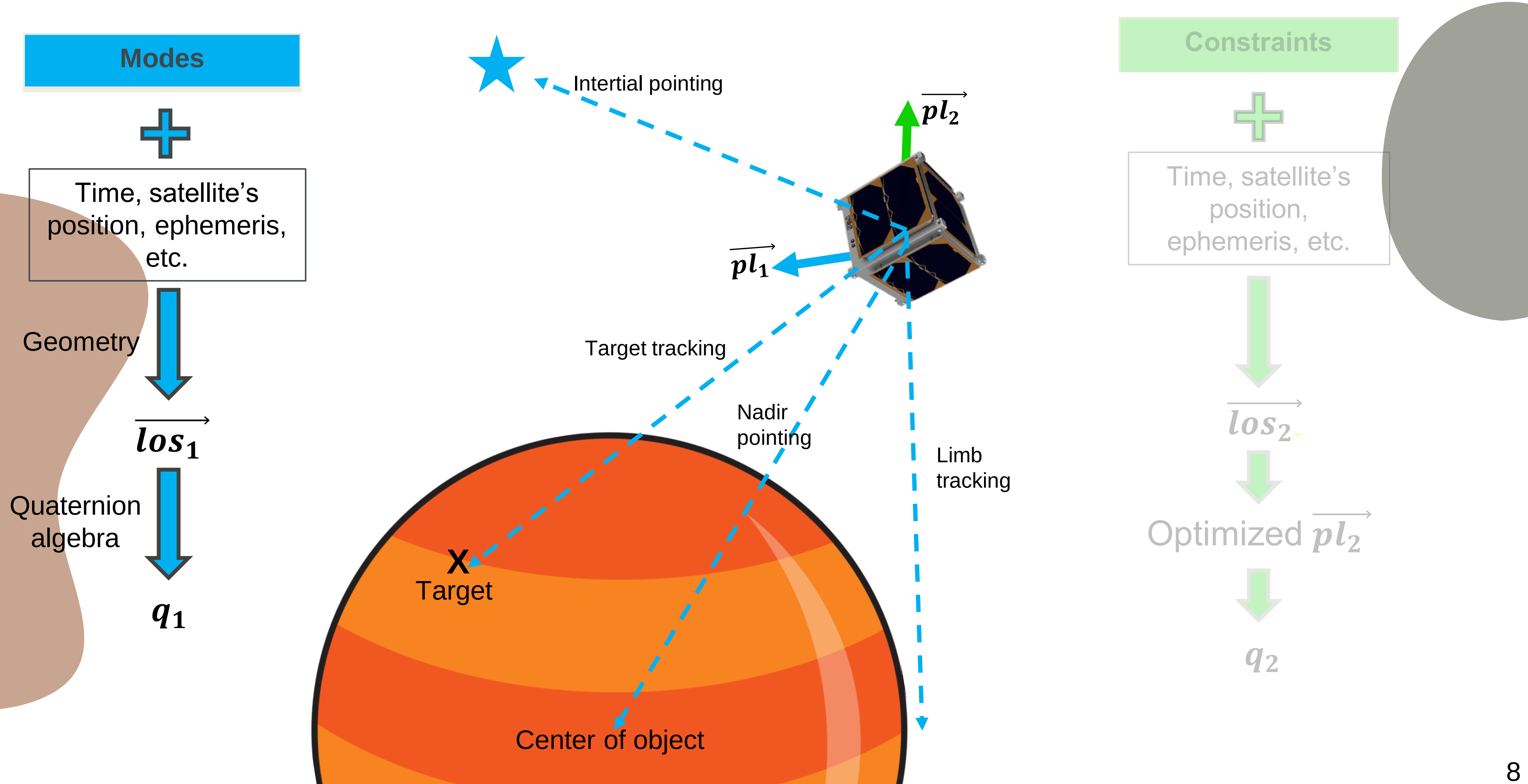
$$\overrightarrow{v'} = \boldsymbol{q} \otimes \vec{\boldsymbol{v}} \otimes \boldsymbol{q}^*$$



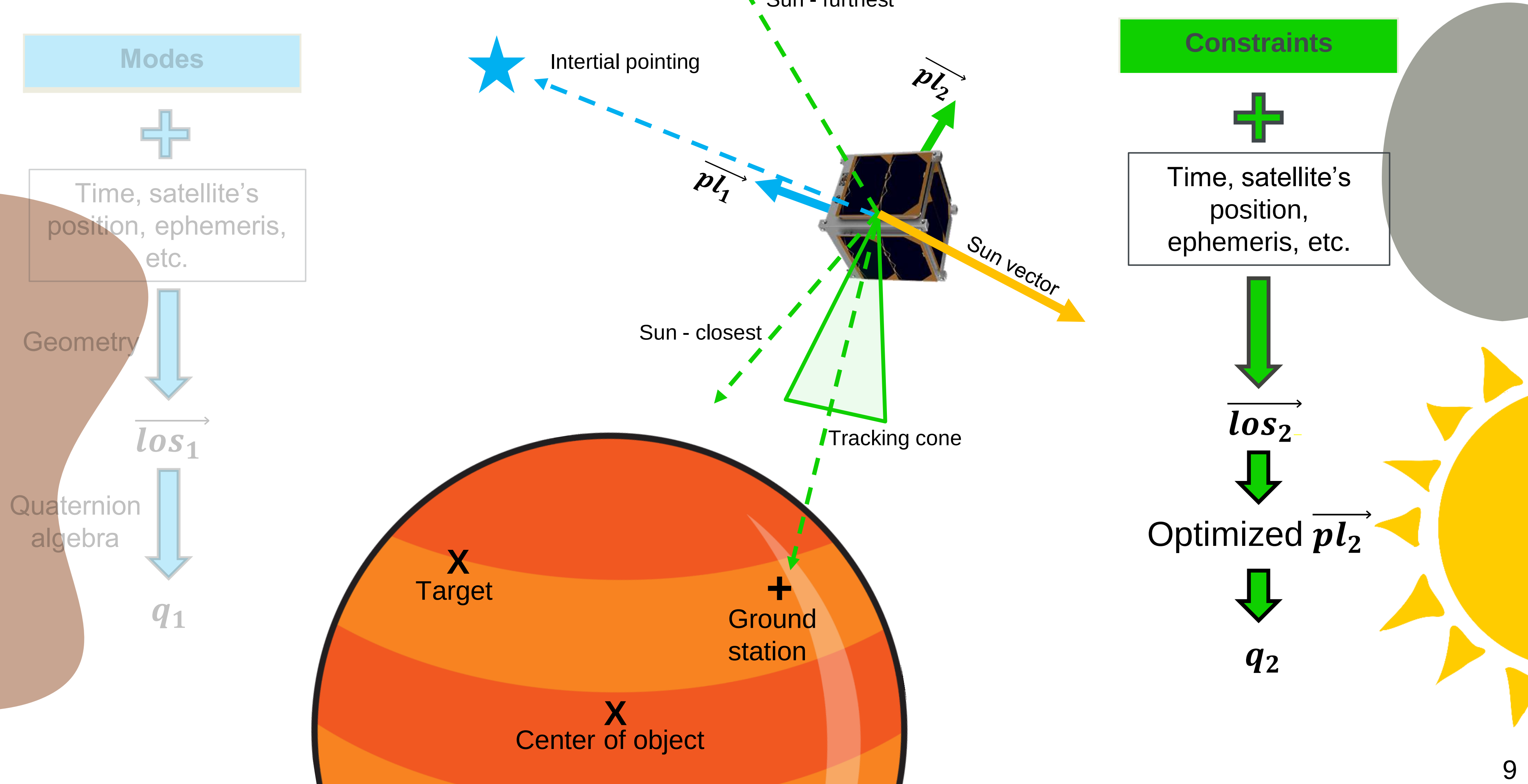


Credit: B. Segret

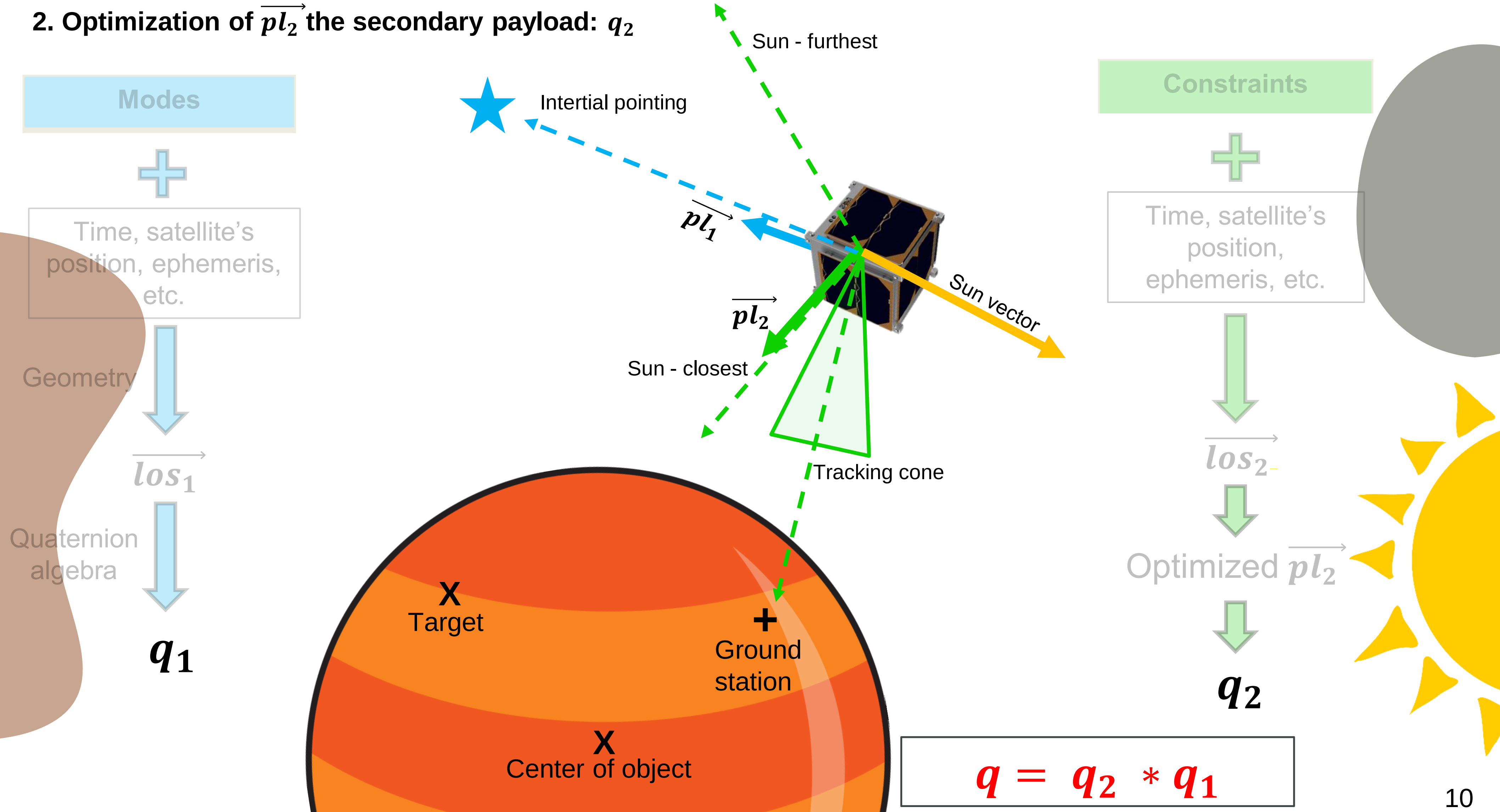
1. Alignment of $\overrightarrow{pl_1}$ the primary payload: q_1



2. Optimization of $\overrightarrow{pl_2}$ the secondary payload: q_2



2. Optimization of $\overrightarrow{pl_2}$ the secondary payload: q_2



$$q = q_2 * q_1$$

IV. Prototype of the module

Modes	
Nadir pointing	Sun pointing
Inertial pointing	Body pointing
Target pointing	Scanning
Limb tracking	User-defined

Constraints	
Sun - optimized	Rotation

```yaml
# ================Trajectory file
traj_file: 'inputs/input_trajectory.txt'
#================Pointing strategies
strategies:
    stg0:
        epoch: '2011-01-01T23:30:00'
        end:
        payload_1_vector : [1,0,1]
        payload_2_vector : [0,-1,0]
        payload_1_mode: 'nadir'
        mode_1_request:
        payload_2_mode: 'rotation'
        mode_2_request: 5087
    stg1:
        epoch: '2011-01-02T23:30:00'
        end: '2011-01-03T00:30:00'
        payload_1_vector : [0,0,1]
        payload_2_vector : [0,0.5,0.866]
        payload_1_mode: 'scan'
        mode_1_request: [20,0]
        payload_2_mode: 'sun_optimized_closest'
        mode_2_request:

#================ outputs directory
File_name: 'output_AEM.txt'
Output_directory: '../Downloads/VTS/Data/DOCKS'
```

IV. Prototype of the module

Modes	
Nadir pointing	Sun pointing
Inertial pointing	Body pointing
Target pointing	Scanning
Limb tracking	User-defined

Constraints	
Sun - optimized	Rotation

```
CIC_AEM_VERS = 1.0
CREATION_DATE = 2023-08-10 17:34:27.359169
ORIGINATOR = CENSUS / LabEx ESEP - Paris Observatory - PSL University Paris

META_START

OBJECT_NAME = DOCKS_Sat
OBJECT_ID = CORPS
REF_FRAME_A = ICRF
REF_FRAME_B = SC_BODY_1
ATTITUDE_DIR = A2B
TIME_SYSTEM = TDB
ATTITUDE_TYPE = QUATERNION
QUATERNION_TYPE = FIRST

META_STOP

59215 0.00000 0.6421100015 0.0000000000 0.0000000000 -0.7666125136
59215 10.00000 0.6421112310 0.0000000000 0.0000000000 -0.7666114838
59215 20.00000 0.6421124583 0.0000000000 0.0000000000 -0.7666104557
59215 30.00000 0.6421136835 0.0000000000 0.0000000000 -0.7666094295
59215 40.00000 0.6421149066 0.0000000000 0.0000000000 -0.7666084051
59215 50.00000 0.6421161275 0.0000000000 0.0000000000 -0.7666073824
59215 60.00000 0.6421173463 0.0000000000 0.0000000000 -0.7666063615
59215 70.00000 0.6421185630 0.0000000000 0.0000000000 -0.7666053424
59215 80.00000 0.6421197775 0.0000000000 0.0000000000 -0.7666043251
```

```
# ================Trajectory file
traj_file: 'inputs/input_trajectory.txt'
#================Pointing strategies
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        payload_1_mode: 'scan'
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#================ outputs directory
File_name: 'output_AEM.txt'
Output_directory: '../Downloads/VTS/Data/DOCKS'
```

Time

Payload vectors

Pointing strategies

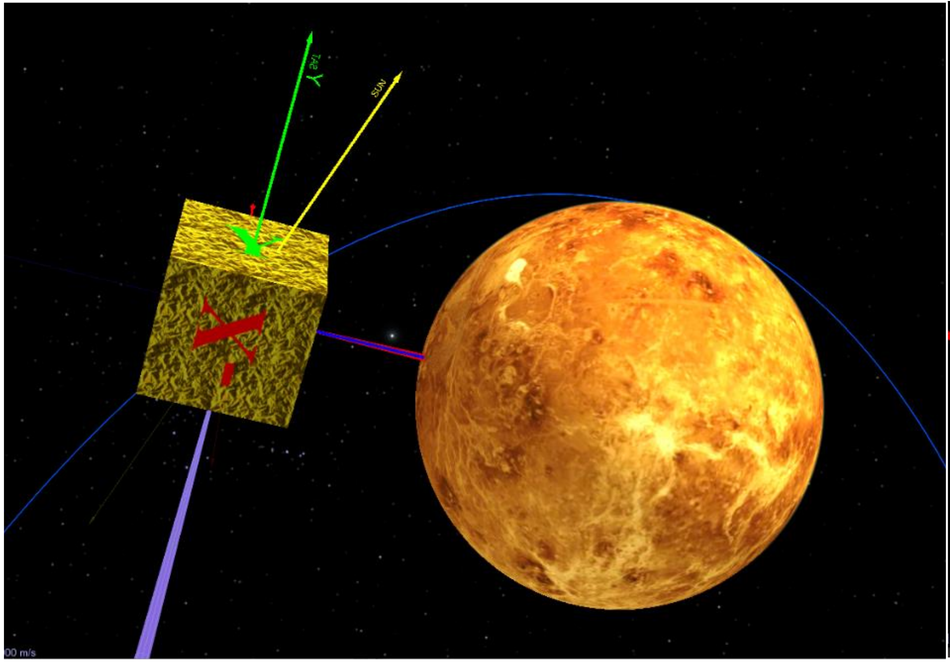
Case study

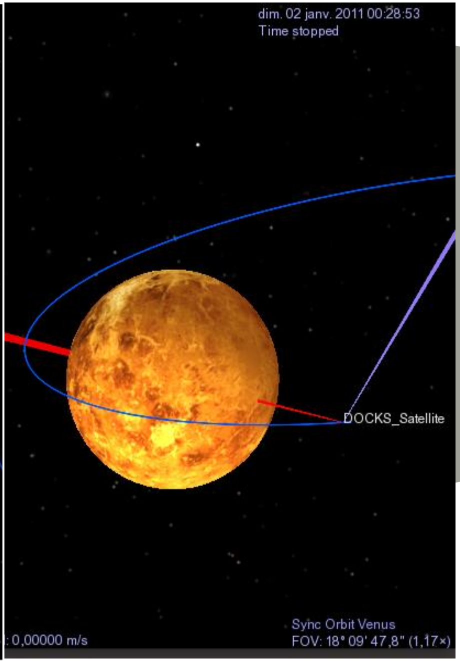
Study case: Venus Express (VEX) with:

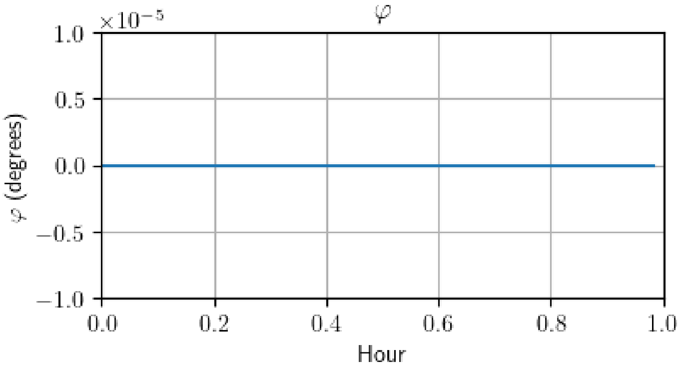
+ $\overrightarrow{\boldsymbol{pl_1}}$ → nadir pointing

+ $\overrightarrow{\boldsymbol{pl_2}}$ → Sun – optimized (to the furthest)

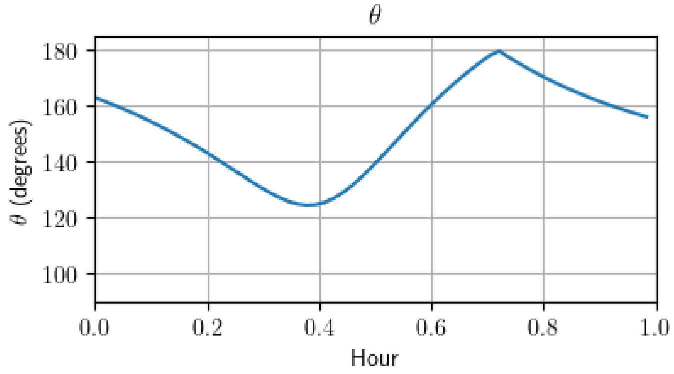
```
epoch: '2011-01-01T23:30:00'
end: '2011-01-02T00:30:00'
payload_1_vector : [0,0,1]
payload_2_vector : [0,-1,0]
payload_1_mode: 'nadir'
mode_1_request:
payload_2_mode: 'sun_optimized_furthest'
mode_2_request:
```







φ: angle between $\overrightarrow{los_1}$ $(\overrightarrow{nadir})$ and $\overrightarrow{pl_1}$ in ICRF



θ: angle between $\overrightarrow{los_2}$ $(\overrightarrow{Sun})$ and $\overrightarrow{pl_2}$ in ICRF

V. Conclusions

1. **Current work**

- Some common and simple pointing strategies have been developed and tested with VTS.
- User's inputs can be loaded to the module by YAML file with multiple parameters.

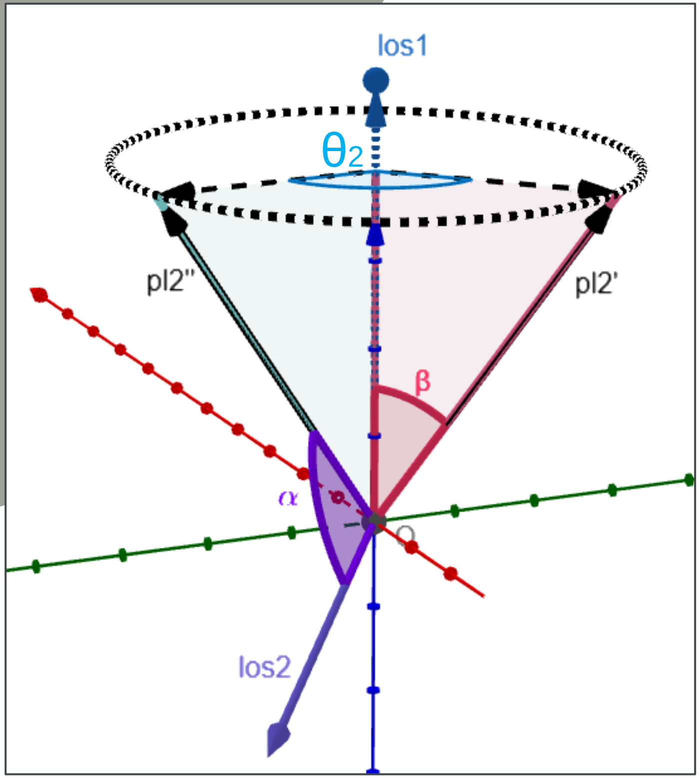
2. **Future work**

- New pointing strategies will be developed and add to the current module.
- New module needs to be integrated with other DOCKS' modules.
- A more effective configuration for the input file is also under investigation

References:

1. CENSUS. https://census.psl.eu/
2. Boris Segret Rashika Jain. *DOCKS, an open-source software suite for space mission profiles.*
3. VTS. Visualization Tool for Space Data. https://timeloop.fr/vts/
4. The SPICE Toolkit. https://naif.jpl.nasa.gov/naif/toolkit.html

Additional slide 1: Optimization of the secondary payload

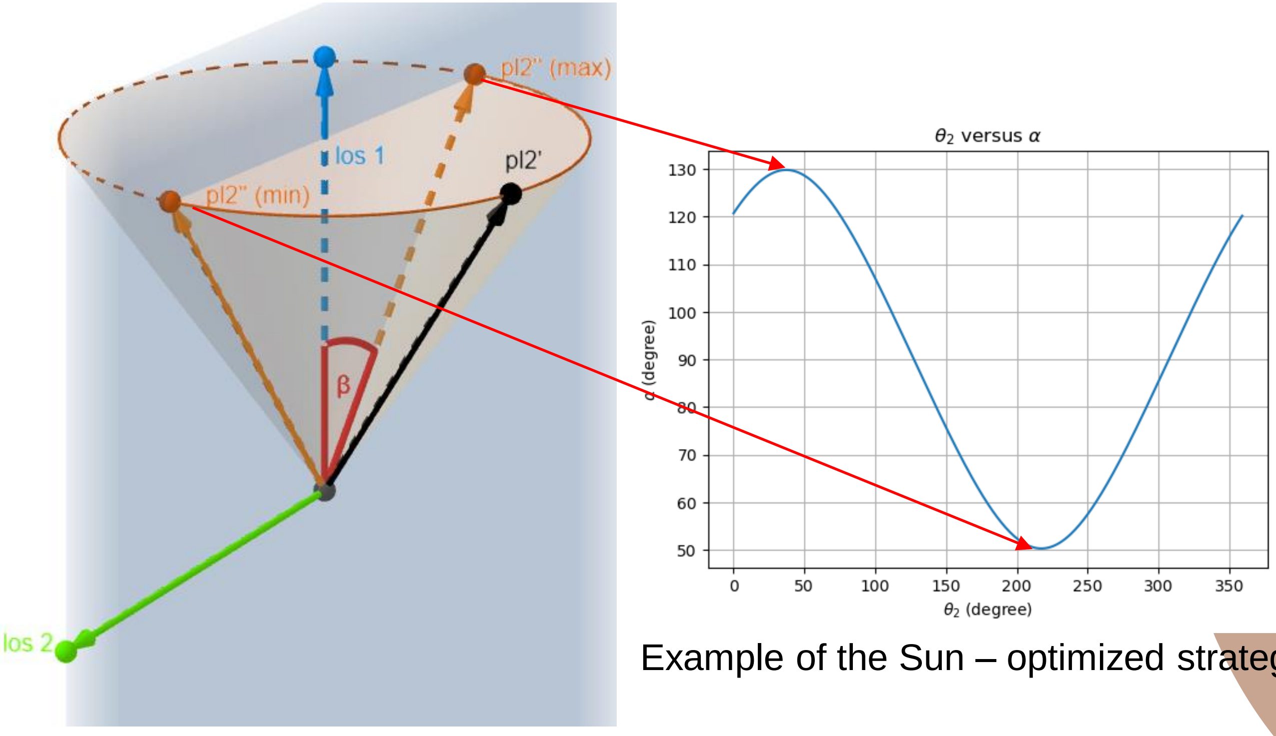


$\overrightarrow{pl2''}$ ≡ optimized coordinate of secondary payload

α: angle($\overrightarrow{pl2''}$, $\overrightarrow{los2}$) ≡ optimized angle with respect to the constraints.

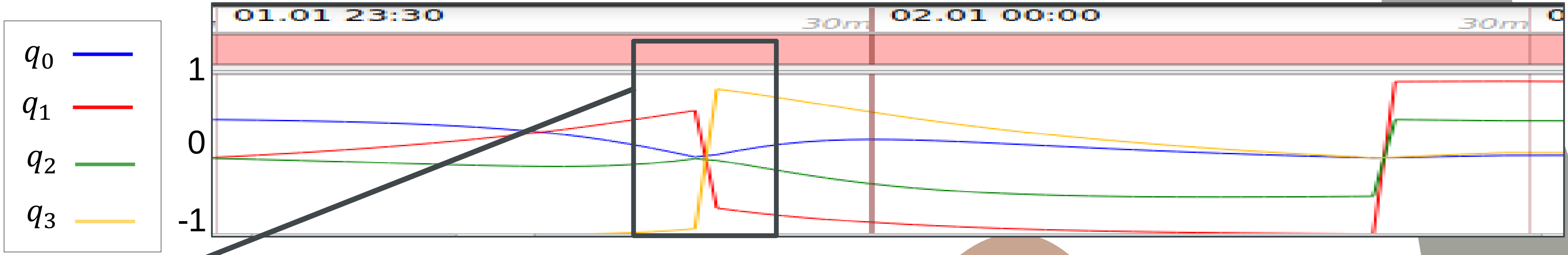
Find α → Find $\overrightarrow{pl2''}$ → create quaternion $\mathrm{q2}$

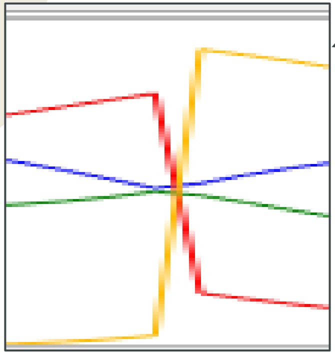
Final quaternion: $\mathrm{q} = \mathrm{q2} * \mathrm{q1}$

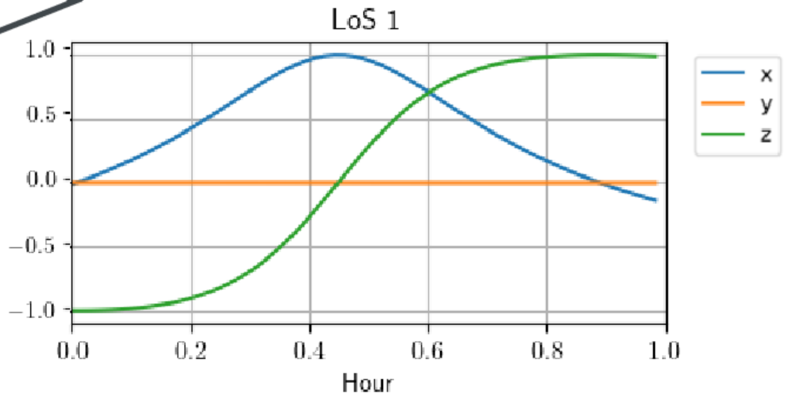


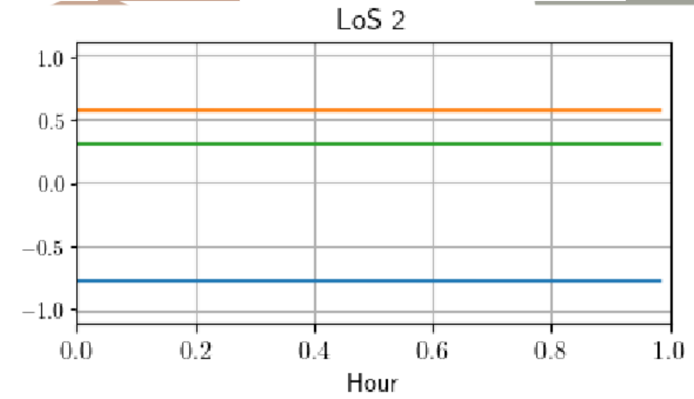
Example of the Sun – optimized strategies

Additional slide 2: Case study investigation



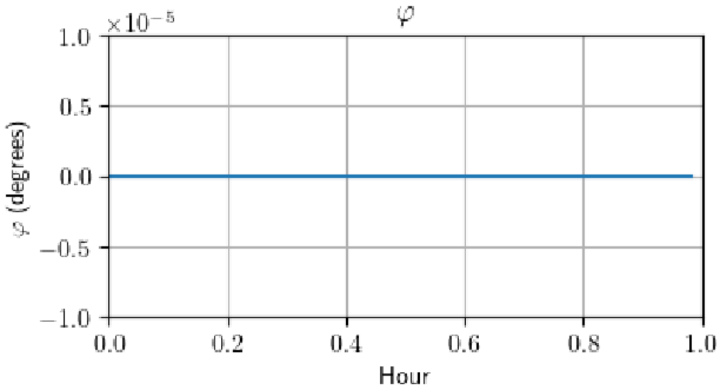


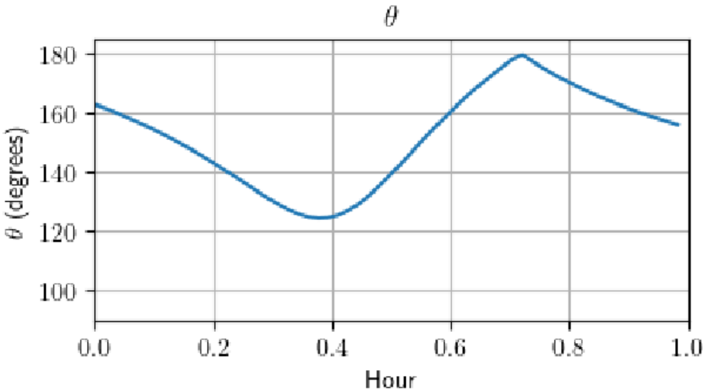




φ: angle between final position $\overrightarrow{pl_1}$ and $\overrightarrow{los_1}$

θ: angle between final position $\overrightarrow{pl_2}$ and $\overrightarrow{los_2}$





Limb

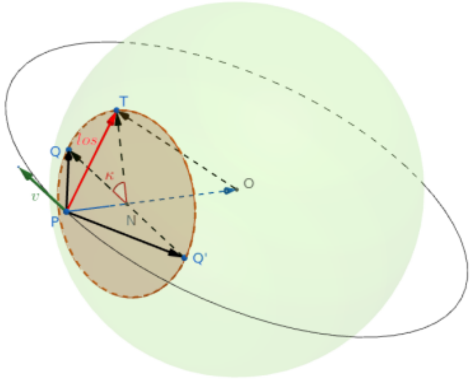
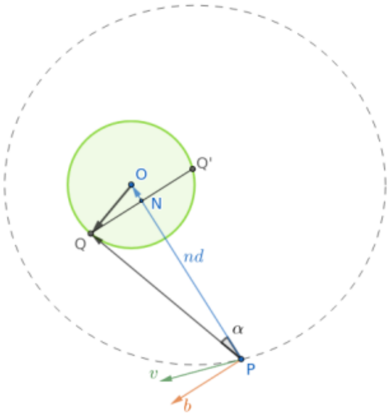


Figure 10: Limb tracking configuration



To find the $\overrightarrow{PQ}$, follow the conventions in Figure 11, the procedure is:

$$\alpha = sin^{-1}(||OT||/||\vec{nd}||)$$

$$\vec{b} = (\vec{nd} \wedge \vec{v}) \wedge \vec{nd}$$

$$\overrightarrow{PQ} = cos(\alpha)\vec{nd} + sin(\alpha)\vec{b}$$

Now quaternion can be created to rotate vector $\overrightarrow{PQ}$ by the angle κ by the axis $\overrightarrow{PO}$, using equation 2:

$$q = (cos(\frac{\kappa}{2}, sin(\frac{\kappa}{2})\frac{\overrightarrow{PO}}{||\overrightarrow{PO}||})$$

$$\overrightarrow{los} = q \otimes \overrightarrow{PQ} \otimes q^*$$

Scan

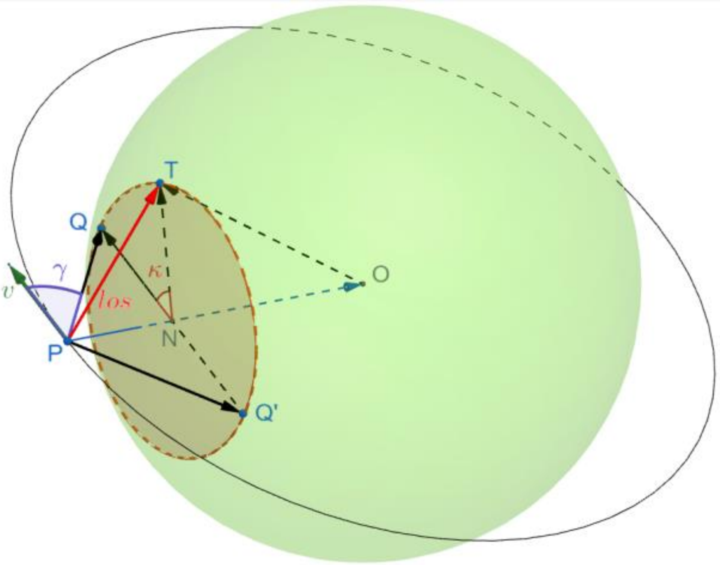


Figure 12: Scan mode configuration

$$\vec{b} = (\vec{v} \wedge \overrightarrow{nd}) \wedge \vec{v}$$

$$\overrightarrow{PQ} = cos(\gamma)\vec{v} + sin(\gamma)\vec{b}$$

$$q = (cos(\frac{\kappa}{2}, sin(\frac{\kappa}{2})\frac{\overrightarrow{b}}{||\overrightarrow{b}||})$$

$$\overrightarrow{los} = q \otimes \overrightarrow{PQ} \otimes q^*$$

Target pointing

$$\overrightarrow{t_{bf}} = \begin{bmatrix} cos(L)cos(\varphi) \\ sin(L)cos(\varphi) \\ sin(\varphi) \end{bmatrix} \Rightarrow \overrightarrow{t_i} = [R_{bf2i}] \cdot \overrightarrow{t_{bf}}$$

$$\overrightarrow{los} = \overrightarrow{OT} - \overrightarrow{p_{sat}}$$

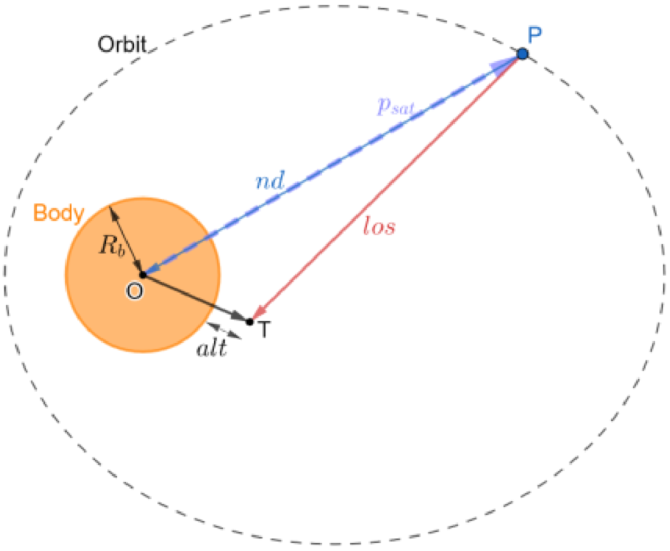


Figure 9: Target pointing configuration

Inertial

$$\vec{e} = \begin{bmatrix} cos(RA)cos(Dec) \\ sin(RA)cos(Dec) \\ sin(Dec) \end{bmatrix}$$

to_body

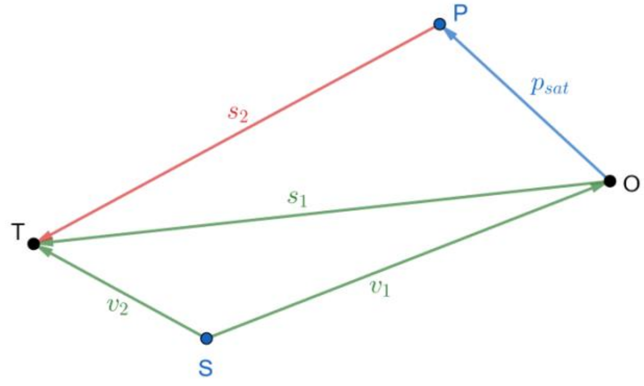


Figure 8: Vectors between bodies in ephemeris